Family 1.3 in two pages

where floating plastic is, and how much — the nets first, the satellites tested against them

blauewelt/earth — two-page summary of the design note, 11 October 2026. A glossary of every term follows the two pages.

What it is. Family 1.3 is a proposed set of datasets ("stores") made of family 1.2 — the programme's global ocean and atmosphere measured at 10 km and 5 days or finer, plus the upper air of the ERA5 reanalysis — **inherited by reference** (listed, not copied), plus a few new stores that say **where floating plastic is in the ocean, and how much**. **What it is for:** a *fifth read-out* beside the programme's four prediction goals (El Niño, the ocean carbon sink, the Atlantic overturning circulation (AMOC) and sea-surface temperature (SST)): floating plastic concentration at a place and a time. It is its own family so that family 1.2's registry and verdicts stay untouched. **What it is not:** not a satellite plastic map — open-ocean plastic is measured almost only by *nets*. A surface net catches pieces down to a third of a millimetre, while the densest open-ocean microplastic covers about 0.005 % of the sea surface (Hu 2021), far below what a 10 m or 1 km pixel can see; satellites see plastic only where it piles up (windrows, river plumes, harbours), and then mixed with seaweed and foam. Not a pile of compilations either — the published tables are mostly the same tows compiled again and again (12,160 of AOMI's 18,860 rows match NCEI on date and position), so each cruise is taken from one producer, the original first. And not built yet: every source was measured on 2026-10-11; nothing has been dispatched, rented or uploaded.

store	what it is	producer	space	time	record	stored size	track	phase
`mplastic`	surface plastic samples — net tows, pump and grab samples, aircraft survey lines; pieces and grams per km^2 *as caught*, the wind-mixing-corrected value beside it, size classes, an empty net as zero	The Ocean Cleanup (Lebreton 2018, 2024); NOAA NCEI; EMODnet; Isobe 2021 once its data licence is confirmed	point; a manta tow sweeps ≈ 0.001–$0.02\,km^2$	per sample	1972-04 → 2025-08	$\approx 20,000$ rows, < 5 MB	public	A
`mobjects`	objects (nets, ropes, buoys, containers . . .) identified in aircraft photo mosaics: length, width, top-view area, weight	The Ocean Cleanup (Lebreton 2018; 2024 if it resolves objects)	point (one object)	per survey	2016; 2021–22 if resolved	1,595 objects, < 1 MB	public	A
`s2labels`	Sentinel-2 pixels labelled by people (marine debris, *Sargassum*, foam, ship, clear or turbid water . . .) with their 11 band reflectances	MARIDA and MADOS (Kikaki 2022, 2024); Plastic Litter Project (Univ. of the Aegean)	one 10–20 m pixel	per scene	2015 → 2023	≈ 2.4 M rows, ≈ 130 MB	public	A
`s2fdi_gpgp`	debris indices (FDI, FAI, NDVI, PI) and their four bands over the Great Pacific Garbage Patch	Copernicus Sentinel-2 level-1C, corrected with ACOLITE	20 m	per overpass (61 dates)	2021-07-02 → 2021-09-19	≈ 1.25 TB before cutting to clear water	public	B, only on a signal
`s2fdi_coast`	the same over coastal waters and enclosed seas	as above	20 m	per overpass	—	a cost estimate, not a build	public	C
`pace_swir`	PACE surface reflectance at six near- and shortwave-infrared wavelengths, 865–2258 nm	NASA PACE OCI (OB.DAAC)	4 km (level 3) or 1.2 km (level 2)	daily	since 2024	≈ 2 TB a year at level 2	public	C
`cygnss_mss`	sea-surface roughness (mean-square slope) and wind, measured from reflected GPS signals	NASA CYGNSS (PO.DAAC)	to measure	to measure	—	to measure	public	C
`beachlitter`	shoreline litter surveys: beach transects and nurdle counts	NOAA NCEI, EMODnet, NOAA MDMAP	point (transect)	per survey; nurdles per 10 minutes	—	≈ 12 k NCEI rows + others	public (NCEI); others to check	C
Inherited from families 1.gf and 1.2 — they bear on plastic but cannot identify it; inherited, not rebuilt								
`oc4k`	daily ocean colour: log-chlorophyll, light attenuation, observation count — the turbid and bloom water a debris index confuses with plastic	family 1.gf	4 km	daily	1997-09 → 2022-12	inherited, not rebuilt	public	—
`pace4k`	PACE colour products: chlorophyll, carbon, plankton cell counts — the same confusers; no reflectance, no shortwave infrared	family 1.gf (NASA PACE)	4 km	daily	2024-03 →	inherited, not rebuilt	public	—
`swot`	sea-surface height and radar backscatter across wide swaths — currents converge where plastic gathers	family 1.gf (SWOT)	2 km	per pass	as in 1.gf	inherited, not rebuilt	public	—
`swh`	wave height along altimeter tracks — waves mix light plastic below a surface net	family 1.gf	along track	along track	as in 1.gf	inherited, not rebuilt	public	—
`icoads`	ship weather reports, here for their winds	family 1.gf	point	per report	as in 1.gf	inherited, not rebuilt	public	—

store	what it is	producer	space	time	record	stored size	track	phase
`era5_u`, `era5_v`	upper-air wind on 13 pressure levels — wind drives both the drift and the mixing	family 1.2 (ERA5)	13 pressure levels	as in 1.2	as in 1.2	inherited, not rebuilt	public	—

Sources and licences. `mplastic` is built from NOAA NCEI's Global Marine Microplastics (CC0 1.0; 16,988 ocean-water samples), The Ocean Cleanup's Lebreton 2018 and 2024 datasets on figshare (CC BY 4.0), EMODnet Chemistry floating micro-litter (CC BY 4.0), and Isobe 2021 (article CC BY 4.0, data licence to confirm); `mobjects` from The Ocean Cleanup (CC BY 4.0); `s2labels` from MARIDA, MADOS and the Plastic Litter Project (all CC BY 4.0, on Zenodo). Sentinel-2 is Copernicus "free, full and open" with the notice "Contains modified Copernicus Sentinel data [year]"; PACE and CYGNSS are NASA open, downloads behind an Earthdata login. **Left out:** AOMI, Japan's Atlas of Ocean Microplastics — non-commercial only, out of phase A, and private if its unique rows prove worth having; Eriksen 2023's "plastic smog" compilation — no licence in its repository (the paper is CC BY 4.0), used only to settle the SEA conflict, which needs no redistribution; CYGNSS's microplastic index — rejected, a roughness anomaly fitted to a plastic model at 1° and 30 days. Sizes are estimates except MARIDA's 837,377 pixels, which were counted.

How it is stored. One registry, `tensors/family1_3/family13.json` (family code 13, version 1.3) on the project's public Hugging Face repository, whose `inherits_block` lists the *resolved* chain — family 1.gf's 13 stores under their own paths and family 1.2's four — so a reader opens one parent list, not a chain (the registry builder needs a small extension for this). The phase-A stores are tier-P point stores built with family 10's point-store builder: one row per net tow, pump or grab sample, aircraft survey line, identified object or labelled pixel, with `qc` carrying the gear class, object type or label class and `platform` the cruise. Values are stored as the net measured them: pieces/km^2 as caught — NCEI's pieces/m^3 is multiplied back by the net depth each compiler assumed, from a committed per-study table — with pumps and grabs per m^3 in their own channel and the producer's wind-mixing correction beside the raw value, never in place of it. A row from a later source is dropped when an earlier one holds the same day within 0.01°, and every drop is counted. Every value carries its footprint: a manta tow its swept area and tow line, a pump sample a point, a Sentinel-2 pixel 20 m.

The ladder

0 Settle the SEA conflict: SEA's 2,529 Pacific tows exist in two copies that agree on monthly counts but match on only 399 positions; compare both with Eriksen 2023's 10,170 SEA rows, or ask SEA. An afternoon.

A `mplastic`, `mobjects`, `s2labels` on a hosted runner, $0 — small enough that the probe IS the build; the per-study depth table; the de-duplication ledger; the registry.

probe One hour, $0: Sentinel-2's only campaign over the patch (summer 2021) coincides with The Ocean Cleanup's 22 aerial surveys; compare the debris index under the survey lines with open water. Expected: nothing above noise (at most $\sim$ 13 objects over 50 cm per km^2, of the order of 0.001 % of the surface).

B `s2fdi_gpgp`, only if the probe showed signal.

C decisions, each with its own probe first: `s2fdi_coast`, `pace_swir`, `cygnss_mss`, `beachlitter`, and AOMI's unique rows on the private track.

Kept only if it earns it. `mplastic` stays a read-out only if a model reading family 1.3's inputs predicts tows held out by cruise better than the five nearest tows and Isobe 2021's climatological map.

Glossary

Every term, acronym, producer, instrument and code name used in this summary and in the full design note, in alphabetical order.

1,732 nm absorption — The wavelength at which plastic absorbs light distinctively; no PACE band sits on it.

5 Gyres — The organisation behind Eriksen's net tows.

accession — An archive's catalogue number for a deposited dataset (NCEI 0306475 for the microplastics table, 0211008 for SEA's own submission).

ACOLITE — An open-source atmospheric correction tuned for water; the debris papers use it on Sentinel-2 level-1C.

adapter — The per-source code that fetches a dataset and converts it into a store (`ml/family1/adapters/...`).

aircraft survey line, aerial survey, photo mosaic — A flight over the sea photographing the surface; the photos stitched into one image, in which floating objects are counted.

altimeter, along track — A radar satellite that measures sea height and wave height straight below it; along-track data lie on that ground line.

AMOC — The Atlantic Meridional Overturning Circulation: the Atlantic currents that carry warm water north near the surface and cold water south at depth; one of the four prediction goals.

anonymous — Readable without any account or login.

AOMI — Japan's Atlas of Ocean Microplastics, 18,860 raw samples; non-commercial only, left out of phase A (D4).

Apache-2.0 — A permissive software licence (FloatingObjects').

ArcGIS FeatureServer, MapServer, Hub — Web map services that answer queries for map records, and the portal that exports them as CSV.

atmospheric correction — Removing the air's contribution from what a satellite sees, to leave the light coming from the surface.

AWS; `sentinel-cogs`, `sentinel-s2-l1c` — Amazon Web Services' cloud storage, and the two public buckets holding Sentinel-2's level-2A and level-1C files.

B04, B06, B08, B11, B12 (bands 4, 6, 8, 11, 12) — Sentinel-2's red, red-edge (740 nm), near-infrared (842 nm) and two shortwave-infrared (1,610 and 2,190 nm) bands.

backscatter (`sig0_karin`) — The share of a radar pulse that bounces back to the satellite; it dims over smooth surface films.

baseline — The simple method a new one must beat to be worth keeping.

`beachlitter` — Phase-C store of shoreline litter surveys: NCEI beach and nurdle counts, EMODnet beach litter, NOAA MDMAP.

Biermann et al. 2020 — The paper that defines the floating debris index (FDI).

bloom — A sudden growth of phytoplankton that colours the water.

carbon sink (ocean) — The ocean's uptake of carbon dioxide from the air; one of the four prediction goals.

CC0, CC BY 4.0, non-commercial — Licences: public domain; free with attribution; free only for non-commercial use.

`chfrank/earth-tensors` — The project's public Hugging Face dataset repository, where family 1.2 is published.

chlorophyll — The green pigment of phytoplankton; its concentration is estimated from ocean colour.

cleanup-system catches — What The Ocean Cleanup's collecting systems brought up (40 in Lebreton 2024).

climatological map, climatology — A long-term average for each place (and season) — the simplest forecast a model has to beat.

CMR — NASA's Common Metadata Repository, the catalogue of its data files.

co-location window, co-location probe — The August 2021 weeks when Sentinel-2 imaged the Great Pacific Garbage Patch while The Ocean Cleanup surveyed it, and the one-hour test that compares the two.

COG — Cloud-optimised GeoTIFF, an image file laid out so a reader can fetch part of it by byte range.

compilation — A table gathered from many studies rather than measured by its publisher.

confusers — Turbid water, blooms, *Sargassum* and foam: what a debris index can mistake for plastic.

Copernicus — The EU's Earth-observation programme, which runs the Sentinel satellites.

CORS `*` — A web header that lets any website read the file from a browser.

crop, patch — A square cut from a satellite scene, here 240 × 240 or 256 × 256 pixels.

cruise — One ship's voyage; tows from the same cruise are held out together so neighbours cannot leak.

CSV — Comma-separated values, a plain-text table.

CYGNSS; `cygnss_mss` — NASA's Cyclone Global Navigation Satellite System, small satellites that measure sea roughness from reflected GPS signals; the phase-C store of that roughness. Its microplastic index (Evans & Ruf 2021) is rejected (D7).

D1–D8 — The eight decisions of the design note: nets first; one producer per cruise; store what the net measured; AOMI out of phase A; Eriksen 2023 out until licensed; no satellite store at scale yet; CYGNSS's index rejected; family code, root and registry.

de-duplication ledger — The record, per pair of sources, of rows read, rows dropped as copies, and rows kept.

derived product — A quantity computed from observations (or a model) rather than measured; admitted only where it carries

observations nothing else does.

DOI — Digital Object Identifier, a permanent link to a dataset or paper (NCEI's: 10.25921/v8gj-k896).

E-085 — The experiment that added ERA5's upper-air fields to family 1.2.

Earthdata login; `EARTHDATA_USERNAME`, `EARTHDATA_PASSWORD` — NASA's free user account, needed to download most NASA data; the repository secrets that hold it.

El Niño — The periodic warming of the equatorial Pacific; one of the four prediction goals.

EMODnet — The European Marine Observation and Data Network; its Chemistry floating micro-litter collection gives raw counts per net.

empty net, zero — A tow that caught nothing; an observation, stored as zero.

ERA5; `era5_u`, `era5_v` — The European weather reanalysis; family 1.2's stores of its eastward and northward upper-air wind on 13 pressure levels.

Eriksen et al. 2014, 2023 — Net tows (implied net depth 0.08 m in NCEI), and the "plastic smog" compilation of 12,546 rows, which has no licence in its repository.

ESA; Sen2Cor — The European Space Agency; its standard Sentinel-2 atmospheric correction, tuned for land, which clips over water.

Evans & Ruf 2021 — The paper behind CYGNSS's microplastic index.

F1 — A score that balances missed detections against false alarms.

FAI — Floating algae index (Hu 2009): band 8 minus a straight line drawn between bands 4 and 11.

falsifier — The test, fixed before building, that can show a store is not worth keeping.

family code, version, root — The registry's identifiers for a family (13, 1.3) and the folder its stores live in (`tensors/family1_3/`).

family; family 1.gf, 1.2, 1.3, 10 — A numbered set of datasets built to one storage design and read by the programme's forecasting model; the global observation stores at 10 km and 5 days or finer; 1.gf plus ERA5's upper air; 1.2 plus the plastic stores; the granularity-aware family whose point-store builder these stores use.

FDI — Floating debris index (Biermann 2020): a Sentinel-2 band difference — band 8 minus a baseline from bands 6 and 11 — that brightens where something floats on the water.

figshare, Zenodo, Springer supplement, GitHub, Google Drive — Public file hosts the sources are served from.

float16 — A number stored in two bytes.

FloatingObjects — Mifdal et al. 2021's line annotations of floating objects on 26 Sentinel-2 scenes (2,270 features).

footprint; `log2_fp`; token — The area and time span a value stands for; its area as a base-2 logarithm, −4 being the label for a point; the unit the model reads, a value with its footprint.

gear class — The kind of sampler: manta, neuston, large net, pump, grab or aircraft; stored in `qc`.

Goldstein — A study of net tows whose implied net depth in NCEI is 0.155–0.25 m.

GPS reflection — Satellite-navigation signals bounced off the sea; their spread measures the surface's roughness.

Great Pacific Garbage Patch (GPGP) — The area of the North Pacific gyre where floating plastic gathers; the box studied is 30–35° N, 135–145° W.

gyre — A basin-wide ring of ocean currents; floating debris collects at its centre.

harmonised table — A compiler's rewrite of many studies into common units, as NCEI's is.

held out by cruise — Left out of training a whole cruise at a time, so a cruise's neighbouring tows cannot reveal the answer.

hosted runner — One of GitHub's own build machines; phase A runs on one, at no cost.

HTTP 200, 206, 302, 401 — Web answers: whole file served; requested byte range served; redirect; login refused.

Hu 2009, Hu 2021 — The floating algae index; the estimate that the densest open-ocean microplastic covers about 0.005 % of the surface.

Hugging Face — The dataset host where the project publishes its stores.

hyperspectral — Resolving a whole spectrum in many narrow bands rather than a handful of broad ones.

`icoads` — Family 1.gf's store of ship and buoy weather reports (ICOADS, the International Comprehensive Ocean-Atmosphere Data Set).

index (spectral) — A formula combining a few bands so that one kind of surface stands out.

inherited by reference; `inherits_block`; resolved chain — A parent's stores listed in the child's registry without a byte copied; the registry field holding that list; the list written out in full, so a reader opens one list and not a chain of parents.

Isobe et al. 2021 — The multilevel dataset (8,218 samples) behind Japan's Atlas of Ocean Microplastics; level 0 and 1 per volume, level 2 per area and wind-corrected, level 3 a gridded climatological map. Its data licence is to be confirmed.

item, tile, scene — One catalogued Sentinel-2 image file; its fixed grid square; one acquisition.

JPEG 2000 (JP2) — The image format of Sentinel-2's level-1C files.

k **nearest neighbours** — A forecast that averages the k closest observations; here the five nearest tows in space and time.

`kd_490` — The attenuation of light at 490 nm, a measure of water clarity.

Kikaki et al. 2022, 2024 — The MARIDA and MADOS labelled sets.

labelled pixels — Satellite pixels a person has marked with what they show; the material any detection method is tested on.

Law et al. 2014 — SEA's 2,529 Pacific tows, whose two published copies disagree (the SEA conflict).

Lebreton et al. 2018, 2024 — The Ocean Cleanup's survey of the Great Pacific Garbage Patch (683 events, 1,595 aircraft objects, 3,532 compiled tows) and its 2015–2022 follow-up (917 manta tows, 162 large-net tows, 74 aerial surveys).

level 1C, 2A; levels 1–4 — Sentinel-2's top-of-atmosphere and atmospherically corrected products; elsewhere, satellite processing stages: calibrated signals, swath measurements, gridded maps, model-filled products.

light attenuation — How quickly light dims with depth in the water; high in turbid or bloom water.

$\log_{10}(1 + \textbf{pieces/km}^2)$ — The scale on which the read-out's error is scored: logarithmic, so a factor of ten counts the same at any density, with 1 added so an empty net scores 0.

LP DAAC, GES DISC, OB.DAAC, PO.DAAC — NASA's land, Earth-science, ocean-biology and ocean data centres.

MADOS — Kikaki 2024's labelled Sentinel-2 set that adds oil spills: 174 scenes, 2,803 crops, CC BY 4.0.

manta net, neuston net, large net — Surface nets that skim the top few tens of centimetres of the sea and so measure pieces per AREA; a bigger net towed for larger pieces.

MARIDA — The Marine Debris Archive (Kikaki 2022): 837,377 human-labelled Sentinel-2 pixels in 15 classes, 3,399 of them marine debris, CC BY 4.0.

mean-square slope — The roughness of the sea surface that CYGNSS measures.

megaplastic, microplastic — Objects over 50 cm; the smallest size class, pieces of 0.05–0.5 cm.

mesh (`mesh_um`) — The size of a net's openings, in micrometres (μm).

Mifdal et al. 2021 — The FloatingObjects annotations.

`ml/build_family1_registry.py` — The script that writes a family-1 registry; it must learn to resolve a chain of two parents (D8).

`mobjects` — Phase-A store of objects identified in The Ocean Cleanup's aircraft photo mosaics.

`mplastic` — Phase-A store of surface plastic samples; the family's ground truth. Channels: `n_km2` (pieces/km^2 as caught), `g_km2` (grams/km^2), `n_m3` (pieces/m^3, volumetric samples only), `n_km2_wc` (wind-mixing-corrected), four size-class counts, `mesh_um`, `depth_m` (net depth), `wind_ms` (wind at sampling).

MSI — The MultiSpectral Instrument on Sentinel-2: 13 bands at 10–20 m.

NASA, NOAA, NCEI — The US space agency; the US National Oceanic and Atmospheric Administration; its National Centers for Environmental Information, the US national environmental archive, whose Global Marine Microplastics table holds 29,076 rows.

NDVI — Normalised difference vegetation index, $(R_8 - R_4)/(R_8 + R_4)$: brightens over floating vegetation.

nm — Nanometre, the unit of wavelength.

NOAA MDMAP — NOAA's Marine Debris Monitoring and Assessment Project of beach surveys.

nurdles — Plastic pellets, the raw material of plastic goods, counted on beaches per 10 minutes.

`oc4k` — Family 1.gf's store of daily 4 km ocean colour.

ocean colour — The colour of light leaving the sea, from which chlorophyll and water clarity are estimated.

OPeNDAP — A protocol for reading subsets of scientific files over the web.

PACE, OCI; `pace4k`, `pace_swir` — NASA's Plankton, Aerosol, Cloud, ocean Ecosystem satellite (2024) and its hyperspectral Ocean Color Instrument; family 1.gf's daily 4 km store of its colour products; the phase-C store of its surface reflectance at six wavelengths.

percentile — The value below which a given share of the data lies.

phase A, B, C — The build waves: A cheap and first, B taken only on a probe's result, C decisions taken later, each with its own probe.

phytoplankton, plankton — Microscopic drifting algae (and the drifting life they feed); PACE counts three groups of their cells.

PI — Plastic index (Themistocleous 2020), $R_8/(R_8 + R_4)$.

Plastic Litter Project — The University of the Aegean's KNOWN plastic targets moored in the Gulf of Gera, Lesvos, imaged by Sentinel-2 and drones (Topouzelis), CC BY 4.0.

`platform`, `qc` — A point store's instrument column (here the cruise) and its quality-or-class column (here gear class, object type or label class).

point-store builder — Family 10's code that writes a tier-P store; family 1.3's phase-A adapters feed it.

pressure level — A height in the atmosphere named by its air pressure (ERA5 uses 13 of them).

probe — A test run of a store's real fetch code before any full build; for phase A's small stores the probe is the build.

proxy — The network gateway the project machine's web requests pass through; some answers may differ outside it.

public track, private track — A store anyone may read from the public repository, or one held for this project's own use because its licence forbids republishing.

pump sample, grab sample — Water pumped or scooped up and filtered — a measurement per VOLUME, kept in pieces/m^3.

$\mathbf{R}_n$, λ_n — The reflectance in Sentinel-2 band n, and that band's centre wavelength.

range read — A web request for one byte span of a file rather than the whole of it.

read-out — A quantity the model is asked to predict or report; family 1.3's is floating plastic concentration.

reanalysis — A weather or ocean model run over past decades and continually corrected towards observations.

red edge, NIR, SWIR — Light just beyond red; near infrared; shortwave infrared.

reflectance; SFREFL, `rhos`; water-leaving reflectance — The share of sunlight a surface sends back at each wavelength; PACE's surface-reflectance product and its variable; the reflectance of the water itself, which in PACE stops at 719 nm.

registry; `family13.json` — The JSON file listing every store of a family; family 1.3's, at `tensors/family1_3/`.

requester-pays — A storage bucket whose downloads are billed to the reader.

river plume — River water spreading into the sea, often carrying debris.

`s2fdi_coast`, `s2fdi_gpgp` — Possible stores of Sentinel-2 debris indices over coastal waters (phase C) and over the Great Pacific Garbage Patch in 2021 (phase B, only if the probe shows signal).

`s2labels` — Phase-A store of human-labelled Sentinel-2 pixels (MARIDA, MADOS, Plastic Litter Project).

Sargassum — A floating brown seaweed, one of the debris index's confusers.

SEA; SEA conflict — The Sea Education Association, whose cruises towed Law 2014's Pacific nets; the disagreement between SEA's own copy of those 2,529 tows and the copy in Lebreton 2018, to be settled before `mplastic` is built.

Sentinel-2, S2A, S2B — The EU Copernicus optical satellites, 10–20 m; the two satellites, whose debris-index factor is 1.7722 and 1.7770.

size classes — The bands of piece size a producer counts separately: 0.05–0.5, 0.5–5, 5–50 and over 50 cm.

SST — Sea-surface temperature; one of the four prediction goals.

store — One dataset of a family, published as a small set of arrays.

surfactants — Natural films that smooth the sea surface and confound CYGNSS's index.

swath — The strip of surface a satellite instrument sees on one pass.

`swh` — Family 1.gf's store of wave height along altimeter tracks.

SWOT; `swot` — The Surface Water and Ocean Topography satellite, which maps sea height across wide swaths; family 1.gf's 2 km store of it.

TB, GB, MB, k, M — Tera-, giga- and megabytes; thousand, million.

The Ocean Cleanup — The organisation whose surveys produced Lebreton 2018 and 2024; first in `mplastic`'s order of precedence.

Themistocleous et al. 2020 — The paper that defines the plastic index (PI).

tier P, G, G-fine — Storage forms: points, profiles and tracks; gridded fields; grids finer than 0.25° stored as compressed tiles.

Topouzelis 2020 — The Plastic Litter Project paper showing detection of targets covering at least 25 % of a pixel.

training split, test split — The labelled data a method is tuned on, and the part kept back to score it.

transect — A fixed stretch of beach walked and counted in a litter survey.

turbid — Clouded by sediment or particles; turbid water is a confuser for debris indices.

verdict — A store's recorded keep-or-drop result.

wind-mixing correction — The producer's scaling up of a net's catch for light pieces that waves pushed below the net; stored beside the raw value.

windrows — Lines on the sea where wind and currents gather floating material.